

Advanced (Carolina Reaper)

- Q1.** If $f(x) = x^2$ and $g(x) = 2x + 1$, what is $f(g(x))$?
(A) $4x^2 + 4x + 1$
(B) $2x^2 + x + 1$
(C) $x^2 + 2x + 1$
(D) $x^2 + 1$
(E) $4x^2 + 1$
- Q2.** Given $f(x) = \frac{1}{x}$ and $g(x) = x - 2$, what is the domain of $f(g(x))$?
(A) $(-\infty, 2) \cup (2, \infty)$
(B) $(-\infty, 0) \cup (0, \infty)$
(C) $(-\infty, 2)$
(D) $(2, \infty)$
(E) $(-\infty, 2) \cup (2, \infty)$
- Q3.** A bakery sells a total of $f(x) = 250 - 5x$ loaves of bread per day, where x is the price in dollars. If $g(x) = x + 2$ represents the price increase, what is the new total number of loaves sold?
(A) $250 - 5(x + 2)$
(B) $250 - 5x + 2$
(C) $250 + 10 - 5x$
(D) $250 - 10 - 5x$
(E) $250 - 5x - 2$
- Q4.** A water tank can hold $f(x) = 1000 - x^2$ gallons of water, where x is the amount of water already in the tank. If $g(x) = x - 5$ represents the amount of water removed, what is $f(g(x))$?
(A) $1000 - (x - 5)^2$
(B) $1000 - x^2 - 5$
(C) $1000 + x^2 - 5$
(D) $1000 - x^2 + 5$
(E) $1000 + (x + 5)^2$
- Q5.** A company has a profit function $f(x) = 200x - x^2$ and a cost function $g(x) = x^2 + 100$. What is the profit after applying the cost function?
(A) $200(x^2 + 100) - (x^2 + 100)^2$
(B) $200x - (x^2 + 100)^2$
(C) $200(x^2 + 100) - x^2$
(D) $200x - x^2 - 100$
(E) $200(x^2 + 100) - x^2 - 100$
- Q6.** If $f(x) = \sqrt{x}$ and $g(x) = x^2 - 4$, what is the domain of $f(g(x))$?
(A) $[-2, 2]$
(B) $(-\infty, -2] \cup [2, \infty)$
(C) $[-2, 2)$
(D) $[2, \infty)$
(E) $(-\infty, -2)$
- Q7.** A person has $f(x) = 2x + 5$ hours of free time per day and spends $g(x) = x - 1$ hours watching TV. How many hours of free time are left after watching TV?
(A) $2(x - 1) + 5$
(B) $2x + 5 - (x - 1)$
(C) $2x + 5 + x - 1$
(D) $2x + 5 - x + 1$
(E) $2x - 5 - x - 1$
- Q8.** If $f(x) = x^3$ and $g(x) = \frac{1}{x}$, what is $f(g(x))$?
(A) $\frac{1}{x^3}$
(B) x^3
(C) $\frac{1}{x}$
(D) x
(E) x^2

Open Ended Questions (FRQ)

- Q9.** A company has a production function $f(x) = 50x - x^2$ and a quality control function $g(x) = x + 2$. Find $f(g(x))$ and determine the domain of $f(g(x))$. Show all steps and explain your reasoning.
- Q10.** A water purification system has a flow rate function $f(x) = 100 - 2x$ and a filtration function $g(x) = x - 1$. Find the composite function $f(g(x))$ and interpret its meaning in the context of the water purification system. Show all steps and explain your reasoning.

Chef's Tips